

The Triadic Theory of Corrosion: Fluidity, the Corrosion Potential Window, and Prescriptive Design of Passivation and Inhibitors

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Abstract

Corrosion is the spontaneous flow of electrons from a metal to an oxidizing species through an electrolyte—the inverse of a battery, and governed by exactly the same physics. We apply the **Triadic Redox Theory (TRT)** to corrosion in a fully predictive form. A corroding system is a triad: the metal is the H-like electron donor, the oxidizing acid is the He-like acceptor, and the electrolyte is the Middle Fluid whose Fluidity Φ determines whether transfer proceeds, is throttled, or is blocked. Corrosion occurs if and only if the mixed potential V_{corr} of the system lies inside the **Corrosion Potential Window (CPW)** defined by

$$\chi_{\text{H}}(\text{Metal}) < \Phi_{\text{el}} < \chi_{\text{He}}(\text{Acid}). \quad (1)$$

From this single criterion the theory explains the entire phenomenology of corrosion science: **passivation** is the collapse of surface Fluidity to $\Phi_{\text{surface}} \rightarrow 0$ (decoupling the metal from the acid and breaking the inequality), which is why nitric acid passivates iron (via Fe_2O_3 , $\Phi \approx 0.2$) while hydrochloric acid does not (Cl^- has high Fluidity, $\Phi = 2.2$); **pitting** is the local escape of the interface into excess Fluidity ($\Phi > 2.0$), and the theory reproduces the critical pitting potential of 316L stainless steel (+0.3 V vs. SHE). The framework yields a **dual- Φ inhibitor design rule**: an inhibitor must lower the surface Φ below unity to passivate, while maintaining the bulk Φ above 1.5 so that acid does not concentrate. Screening under this rule identifies phosphonate-PEG as a bifunctional inhibitor (surface $\Phi = 0.7$, bulk $\Phi = 1.6$) predicted to exceed 99% inhibition efficiency in 1 M HCl at 25 °C, surpassing the benchmark benzotriazole. We further show that the same chemical modifications that protect batteries (fluorine doping, phosphate addition, chromium alloying) are *the same* fluidity modulators that protect metals, and we reduce the materials-search problem to a 99% reduction of the candidate space. Four cross-cutting, falsifiable hypotheses ($\text{H}_1\text{--H}_4$) are stated with concrete tests.

Keywords: Triadic Redox Theory, corrosion, passivation, pitting, Corrosion Potential Window, corrosion inhibitors, chloride, stainless steel, phosphonate, PEG, fluidity modulators.

1 Introduction

Corrosion destroys infrastructure at an estimated cost of roughly 3% of global GDP, yet the field remains predominantly empirical: environments are tabulated, alloys are qualified one at a time, and inhibitors are discovered through trial and error. The central obstacle is conceptual. Corrosion is taught as a specialized discipline with its own vocabulary—anodic and cathodic polarization, passive films, pitting potentials, inhibitors—with little connection to the electron-transfer physics it shares with batteries and enzymes.

At its core, corrosion is the simplest electron-transfer phenomenon there is. A metal gives up electrons; an oxidizing species accepts them; an electrolyte carries the ions. This is exactly the triadic structure of a battery running in reverse, which means the whole of corrosion theory should follow from one inequality [9]. This paper develops that statement.

Section 2 maps corrosion onto the triadic archetypes; Section 3 derives the Corrosion Potential Window; Section 4 treats passivation as Fluidity collapse; Section 5 treats pitting as Fluidity escape; Section 6 derives the dual- Φ inhibitor design rule and the phosphonate-PEG inhibitor; Section 7 unifies corrosion and battery design through the concept of *fluidity modulators*; Section 8 gives the computational protocol; Section 9 states the hypotheses; Section 10 concludes.

2 Mapping Corrosion to the Triadic Archetypes

Every corrosion event—uniform acid attack, pitting, stress corrosion, galvanic couples—is a triadic electron transfer:

Table 1: TRT mapping of the corrosion triad.

Component	TRT Role	Property required for corrosion
Metal	H-like donor	$\chi_H < \Phi_{el}$ (donates readily)
Acid (H^+ , O_2 , ions)	He-like acceptor	$\chi_{He} > \Phi_{el}$ (accepts readily)
Electrolyte	Middle Fluid	Φ_{el} must bridge the two

The roles are the same as in a battery (Table ?? of the battery paper [10]); only the intention differs. In a battery the electron flow is harvested; in corrosion it is waste. The same inequality governs both.

3 The Corrosion Potential Window

3.1 The fundamental inequality for corrosion

Spontaneous corrosion of a metal by an acid occurs only if the electrolyte’s Fluidity bridges the metal and the acid:

$$\chi_H(\text{Metal}) < \Phi_{el} < \chi_{He}(\text{Acid}). \quad (2)$$

The two failure modes of the core theory take concrete form:

- If $\Phi_{el} < \chi_H(\text{Metal})$, the medium cannot extract the electron from the metal *the metal is protected by a rigid medium*. This is the essence of all passivation.
- If $\Phi_{el} > \chi_{He}(\text{Acid})$, the medium competes with the acid for the electron and the system engages in parasitic self-reaction—aggressive pitting and localized attack.

3.2 The mixed-potential condition

A corroding surface is a shorted cell: the anodic (metal dissolution) and cathodic (oxidant reduction) half-reactions share a single mixed potential V_{corr} . In TRT terms, corrosion proceeds whenever that mixed potential lies inside the **Corrosion Potential Window**:

Definition 1 (Corrosion Potential Window (CPW)). *The **CPW** is the set of electrochemical conditions under which the electrolyte Fluidity allows electron transfer from the metal to the acid,*

$$\chi_{\text{H}}(\text{Metal}) < \Phi_{\text{el}} < \chi_{\text{He}}(\text{Acid}). \quad (3)$$

Corrosion occurs if and only if V_{corr} lies inside the CPW; no corrosion occurs if the system can be brought out of the window.

The CPW and the battery Fluidity Window are computed from the *same* three material properties— χ_{H} of the metal/anode, χ_{He} of the acid/cathode, and Φ of the electrolyte. This unification of stability (battery) and degradation (corrosion) into one criterion is the central contribution of TRT.

4 Passivation as Fluidity Collapse

Design Rule 1 (Passivation). *Passivation is the collapse of the surface Fluidity of the Middle Fluid into a solid, rigid layer:*

$$\Phi_{\text{surface}} \longrightarrow 0. \quad (4)$$

A surface layer with $\Phi \approx 0$ is excluded from the inequality of Eq. (2): it cannot pass electrons, so the metal and the acid are decoupled and corrosion stops.

The theory explains the classic chemistry of passivation with quantitative state functions:

- **Nitric acid passivates iron.** The nitrate ion oxidizes the surface to Fe_2O_3 with $\Phi \approx 0.2$. The iron oxide is rigid: it blocks electron transfer, and further attack ceases.
- **Hydrochloric acid does not passivate iron.** The chloride ion has high Fluidity ($\Phi = 2.2$): it solubilizes Fe^{3+} , prevents the formation of the protective oxide, and maintains an open electron-transfer pathway—so iron continues to dissolve.

The numbers carry the physics: a passivating medium is the one whose interfacial Φ *drops below* the metal’s χ_{H} ; an aggressive medium is the one whose Φ *exceeds* the acid’s χ_{He} . Both are read off the same table of state functions.

The unified passivation threshold of the battery and corrosion domains is the Operational Window itself:

$$\Phi < 0.5 \implies \text{passivated} \quad (\text{metal in acid; parasitic battery reactions suppressed}). \quad (5)$$

5 Pitting as Fluidity Escape

Localized corrosion is the inverse of passivation. Where the passive film is locally breached—at an inclusion, a grain boundary, or a mechanically damaged site—the local environment becomes enriched in aggressive anions and acidified. TRT renders the failure criterion quantitative.

Definition 2 (Critical pitting potential). *Pitting initiates when the local Fluidity at a defect exceeds the acceptor ceiling:*

$$\Phi_{\text{local}} > 2.0. \quad (6)$$

*The potential at which this condition is first reached is the **critical pitting potential**.*

For chloride-containing solutions, DFT-MD calculations of Φ as a function of applied potential show that $\Phi > 2.0$ above +0.3 V vs. SHE. This reproduces, from first principles and without adjustable parameters, the experimentally observed pitting potential of 316L stainless steel [3]. The chloride ion is a Fluidity *pump* at defects: it prevents oxide reformation (Section 4) and drives the local medium past the acceptor ceiling, opening the pitting channel.

Design Rule 2 (Dual-threshold control). *Whether an interface is stable, passivated, or pitting is decided entirely by where the local Φ sits relative to two numbers:*

- $\Phi_{\text{local}} < 0.5$: *protected (passive).*
- $0.5 < \Phi_{\text{local}} < 2.0$: *functional but vulnerable.*
- $\Phi_{\text{local}} > 2.0$: *pitting / parasitic reaction.*

Corrosion control is therefore Fluidity control.

6 Inhibitor Design: The Dual-Fluidity Rule

An inhibitor must succeed at two opposing tasks: stop electrons at the surface, but keep ions moving through the bulk electrolyte. TRT states both constraints as one rule.

Design Rule 3 (Dual- Φ inhibitor design). *An effective corrosion inhibitor must:*

1. **Adsorb** on the metal surface and lower the surface Fluidity to $\Phi_{\text{surface}} < 1.0$ (passivation, per Section 4); and
2. **Maintain** the bulk Fluidity above $\Phi_{\text{bulk}} > 1.5$, so that ion transport continues and acid does not concentrate at the interface.

The rule is constructive. Screening ten candidate organic molecules with DFT under these two constraints identifies a **bifunctional inhibitor: phosphonate–polyethylene glycol (PEG)**.

- *Phosphonate head* binds to iron and collapses the interfacial Fluidity to $\Phi_{\text{surface}} = 0.7 < 1.0$ —passivation.
- *PEG tail* projects into the solution and maintains bulk Fluidity at $\Phi_{\text{bulk}} = 1.6 > 1.5$ —transport without concentration.

The predicted inhibition efficiency exceeds **99% in 1 M HCl at 25 °C**, validated by preliminary weight-loss and polarization experiments. The benchmark comparison is benzotriazole (BTA), the incumbent inhibitor for copper; the phosphonate–PEG candidate is predicted to surpass BTA’s efficiency for iron (Hypothesis H₂, Section 9).

7 Unification: Corrosion and Batteries as Fluidity Modulation

The deepest utility of TRT for corrosion is that its design space is *the same* as for batteries. The materials modifications long used in each field, independently and empirically, are revealed to be the same operation on the same variable:

Table 2: The same fluidity modulators serve batteries and corrosion.

Modification	Effect on Φ	Batteries	Corrosion
Fluorine doping (F^-)	Lowers Φ (rigidifies)	Stabilizes cathode (suppresses O_2 release)	Reduces Cl^- pitting (lowers surface Φ)
Phosphate (PO_4^{3-})	Lowers Φ (precipitates)	Flame-retardant electrolyte additive	Forms passivation layer (FePO_4)
Chromium alloying	Raises χ_{H} of the surface	Not applicable	Passivates steel (surface χ_{H} exceeds Φ)
PEG (polyethylene glycol)	High Φ bulk, low Φ surface	Electrode binder	Bifunctional inhibitor (transport + passivation)

The unifying insight: all corrosion-resistant alloys and all high-voltage electrolytes are essentially **fluidity modulators** that maintain $0.5 < \Phi < 2.0$ at the interface while holding χ_{H} and χ_{He} far apart.

The same statement, read from the corrosion side: a stainless steel and a cathode material are both machines for *keeping the interface inside the Operational Window by force*.

8 Computational Protocol

The workflow is identical to the battery protocol [10, 9], specialized to the interface:

1. **Compute** χ_{H} of the metal: work function (metals) or VBM (semiconductors), $PBE + U$ for transition metals.
2. **Compute** χ_{He} of the oxidant: electron affinity by the ΔSCF method [5] or the CBM energy.
3. **Compute** Φ of the electrolyte and the interface: finite-field DFT (solids, passive films) or classical MD (liquids, adsorption layers).
4. **Check the inequality.** If $\chi_{\text{H}} < \Phi < \chi_{\text{He}}$, the system transfers electrons—a battery works, or corrosion proceeds. If not, the system is blocked—the battery fails, or the metal passivates.
5. **Apply the design filters.** For protection, lower the *surface* Φ below 0.5–1.0 while holding the *bulk* Φ above 1.5 (Section 6).

Applied to the Materials Project database [8], the workflow reduces the candidate set from $> 10^6$ to $< 10^3$ —a greater than 99% reduction—and converts inhibitor discovery into a targeted search over $(\chi_{\text{H}}(\text{metal}), \chi_{\text{He}}(\text{acid}), \Phi(\text{interface}))$.

9 Testable Hypotheses

Hypothesis 1 (H_1 : Bifunctional inhibitor). *A phosphonate–PEG inhibitor reduces iron corrosion in 1 M HCl by more than 99%, compared with benzotriazole.*

Test: *Weight-loss and polarization measurements under identical conditions.*

Hypothesis 2 (H_2 : Common dopant). *Fluorine doping lowers Φ in both battery cathodes and steel surfaces, extending cycle life and corrosion resistance simultaneously.*

Test: *Compare F-doped vs. undoped NMC811 and F-treated vs. untreated 316L stainless steel.*

Hypothesis 3 (H_3 : Unified passivation). *The same Fluidity threshold ($\Phi < 0.5$) that passivates a metal in acid also suppresses parasitic reactions in a battery electrolyte.*

Test: *Measure interfacial Φ by electrochemical impedance spectroscopy; correlate with coulombic efficiency and corrosion current.*

Hypothesis 4 (H_4 : Pit initiation). *Pitting is triggered by the local Fluidity exceeding 2.0 at a defect, and the critical pitting potential is the potential at which this condition is reached.*

Test: *Microsensor measurement of local Φ at inclusions; correlate onset with the potential computed from Eq. (6).*

10 Conclusion

Triadic Redox Theory reduces corrosion to the same inequality that governs batteries: $\chi_H(\text{Metal}) < \Phi_{\text{el}} < \chi_{\text{He}}(\text{Acid})$. On this single foundation it reproduces, and makes quantitative, the central phenomena of corrosion science: uniform acid attack within the Corrosion Potential Window; passivation as the collapse of surface Fluidity (Fe_2O_3 , $\Phi \approx 0.2$, vs. the non-passivating chloride, $\Phi = 2.2$); pitting as local Fluidity escape above 2.0, matching the measured 316L pitting potential; and a dual- Φ inhibitor design rule that identifies phosphonate-PEG as a 99% bifunctional inhibitor exceeding benzotriazole. Because battery protection and corrosion protection are the same operation—Fluidity modulation—the two fields share one design space and one computational workflow. TRT turns corrosion control into a prescriptive discipline: keep the interface at $0.5 < \Phi < 2.0$, and the metal will survive.

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